

Volatility Surface Construction and Regime-Gated Directional Forecasting for NIFTY 50 Index Options

Abstract

This report documents an end-to-end quantitative pipeline built on five and a half years (January 2021 – June 2025) of NSE NIFTY 50 index options, futures, and India VIX data, comprising approximately 2.1 million raw option quotes and 386,942 liquidity-filtered records used for implied volatility extraction. The project progresses through eight stages: (1) diagnosing and correcting a systematic pricing bias in a Black-Scholes implied volatility engine via put-call parity, (2) constructing a full volatility smile and term structure surface using a regularized smoothing spline, (3) benchmarking realized volatility, VIX, and regime forecasting against market-implied baselines, (4) incorporating a cross-market spillover signal from U.S. equity markets, (5) building and iteratively refining a regime-gated heuristic that isolates the specific subset of trading days on which directional prediction is reliable, (6) validating that heuristic against a fully data-driven machine learning alternative, (7) comparing four model families (Logistic Regression, Gradient Boosting, Random Forest, XGBoost) to confirm the chosen model is genuinely the best performer rather than an arbitrary choice, and (8) translating the resulting directional signal into a real, capital-tracked options trading backtest. The central finding is that unconditional volatility forecasting cannot beat market-implied baselines – itself an informative result about market efficiency – but that a regime-detection layer built on the confirmation between two independent signals identifies a specific 6–25% subset of trading days on which directional accuracy reaches 62–90%. Applied to a real calendar-spread options strategy with compounding position sizing, this signal produced a net gain of Rs 655,490 (+655.5%) on a Rs 100,000 starting portfolio for the data-driven strategy over the full test period, though with a materially negative final six-month window that is itself informative about the limits of the underlying edge.

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1 Introduction and Motivation

The starting point for this project was a straightforward question: given NSE NIFTY 50 options data, VIX history, and futures data, can implied and realized volatility be predicted with genuine, out-of-sample accuracy – and can any such prediction be turned into a real, capital-tracked trading result? The project evolved through thirteen working notebooks, each building directly on verified output from the previous stage. A recurring design principle throughout is **honest evaluation**: every claimed result is benchmarked against a naive baseline (persistence, market-implied volatility, or a majority-class classifier), evaluated with time-series cross-validation or genuine walk-forward retraining that respects the temporal ordering of the data, and reported alongside its failure modes rather than only its successes. Where an earlier methodological choice was later found to be suboptimal, this report documents the correction and its downstream effect on results, rather than presenting only the final version as if it were reached directly.

1.1 Data Sources

- **NSE NIFTY option chain** (OPTIDX.NIFTY.CE/PE): daily strike-level quotes with Open/High/Low/Close, Jan 2021–Jun 2025, spanning 1,113 trading days.
- **NIFTY futures** (FUTIDX.NIFTY): daily OHLC and underlying index value, used to construct a clean daily spot series.
- **India VIX**: daily OHLC, 1,113 trading days.
- **U.S. equity market data**: CBOE VIX daily closes (retained) and S&P 500 daily closes (tested and ultimately excluded – see Section 6.2).

2 Stage 1: Data Cleaning and Liquidity Filtering

Raw NSE files were ingested and cleaned (date parsing, numeric coercion, handling of ‘-’ placeholder values). A liquidity filter was applied to retain only option quotes suitable for reliable implied volatility extraction (minimum contracts traded, minimum open interest, minimum premium, moneyness range $0.85\text{--}1.15\times$ spot, minimum 2 days to expiry), producing **386,942 liquid option records** from the original raw dataset – the input to all subsequent implied volatility work.

3 Stage 2: Correcting a Systematic IV Pricing Bias

3.1 The Diagnosed Problem

The initial Black-Scholes implied volatility solver used raw spot price with a flat, assumed risk-free rate ($r = 5\%$) for every date and maturity. Because NSE index options are effectively priced off the *futures* curve rather than spot, and the true cost-of-carry varies with both time and the prevailing repo rate, this produced a systematic vertical offset between call-side and put-side implied volatility.

3.2 The Fix: Forward Extraction via Put-Call Parity

The market-implied forward price was recovered directly from the data using put-call parity, fitting $C(K) - P(K) = -e^{-rT}K + e^{-rT}F$ across matched strikes for every (date, expiry) pair. A forward was successfully fitted for **5,284 of 6,724** (date, expiry) groups (1,440 dropped for having fewer

than 3 matched strikes or a poor-quality fit). The recovered forward sat, on average, **0.55% above spot** – a small, sensible cost-of-carry premium, with the full distribution shown in the solver’s diagnostic output. The BSM solver was then re-run using this recovered forward with $r = 0$, solving implied volatility for **378,932 option records**, with a resulting IV distribution of mean 16.6%, median 15.0%, and a 25th–75th percentile range of 12.4%–19.2%.

3.3 Verification

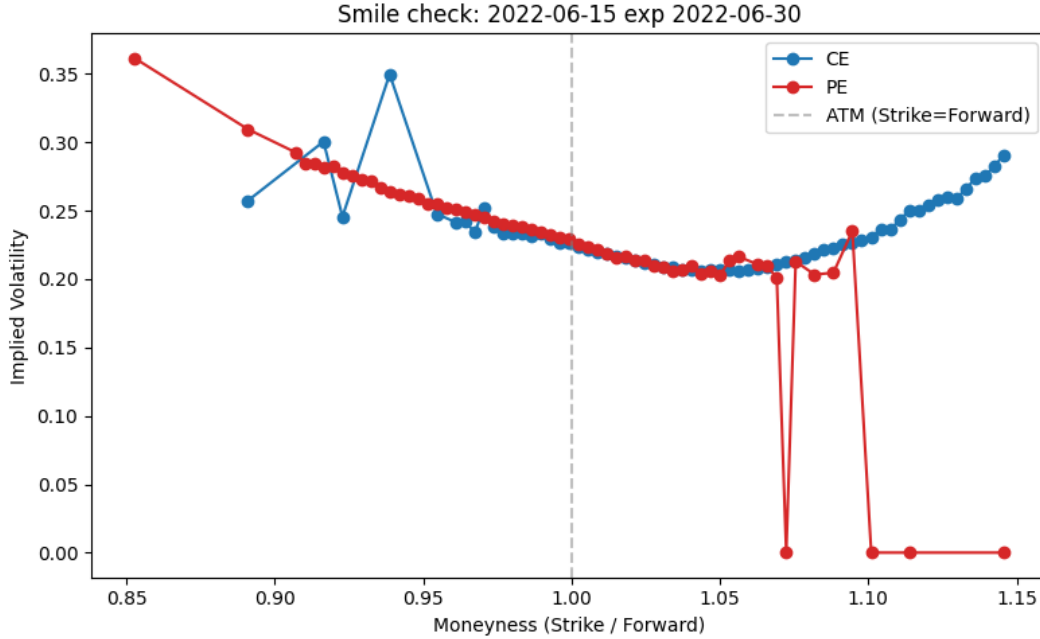


Figure 1: Post-correction smile check for 2022-06-15 (30-day expiry). Call (CE) and put (PE) implied volatilities converge onto a single continuous curve near the at-the-money point (dashed line), rather than sitting on two vertically offset tracks – confirming the forward correction resolved the systematic bias. The visible discontinuities in the deep OTM put wing reflect a small number of illiquid, near-zero-premium quotes rather than a modeling artifact.

4 Stage 3: Volatility Smile and Surface Construction

4.1 Smile Fitting Methodology

Corrected implied volatilities were fit per (date, expiry) as a curve in log-moneyness $k = \log(K/F)$, through three successive iterations, each tested against real sample dates rather than assumed correct.

Iteration 1 – quadratic. An initial fit $IV(k) = a + bk + ck^2$ was found, on visual inspection, to under-fit the curvature of the smile on several sample dates.

Iteration 2 – quartic. A degree-4 polynomial reduced fit RMSE by 5–29% relative to the quadratic, but a structural limitation remained: real NIFTY smiles typically resemble two roughly linear wings meeting a rounded dip near the money, a shape no single global polynomial can represent well everywhere at once.

Iteration 3 – regularized smoothing spline (final). A cubic smoothing spline was fit in place of any polynomial. A first attempt with light smoothing was found, on direct visual inspection, to *overfit* – oscillating between individual noisy quotes despite a numerically lower RMSE than the quartic. The smoothing parameter was corrected to a heavier setting ($s = 3.0 \times n_{\text{points}} \times \text{Var}(\text{IV})$), after which the fitted curve tracked the intended shape cleanly. This is the methodology used in the final pipeline.

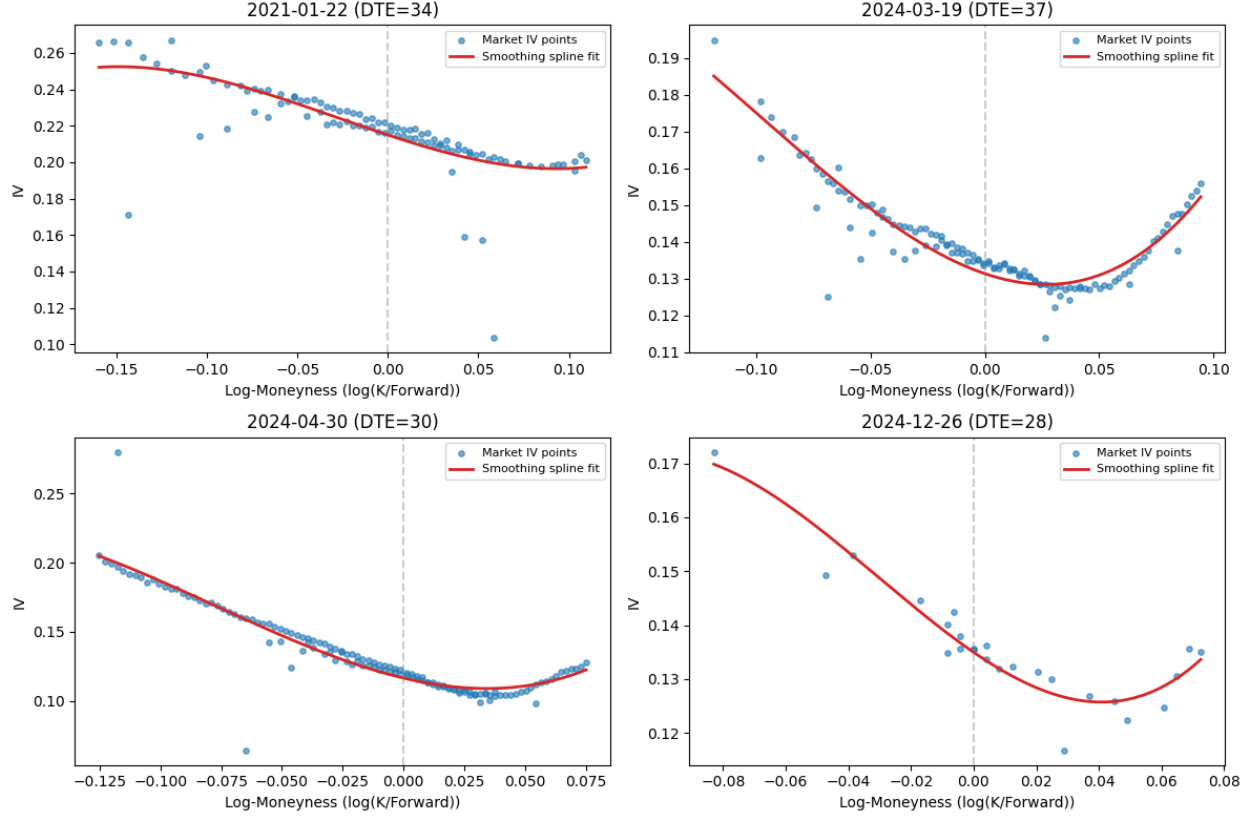


Figure 2: Final spline-fitted smiles on four spot-checked dates. The fitted curve (red) tracks near-linear wing behavior smoothly bending into a rounded ATM dip – the intended shape, without the oscillation seen in an under-smoothed spline or the poor wing fit of a quadratic.

4.2 Multi-Tenor Surface

Smiles were fit and summarized at three tenors – 7, 14, and 30 calendar days. Term-structure slope features (30D – 14D, 14D – 7D, for both skew and curvature) were computed as additional daily features and saved to `smile_features.csv` (25 columns).



Figure 3: Time series of fitted smile level, skew, and curvature at the 30-day tenor across the full sample period. Level visibly tracks known volatility regimes (elevated 2021–2022, calmer 2023, a sharp 2024 spike); skew remains consistently negative, the expected equity-index pattern.

5 Stage 4: Baseline Forecasting Experiments

Three independent unconditional forecasting targets were tested, each benchmarked against the relevant market-implied or naive baseline using Linear Regression and Gradient Boosting.

5.1 Realized Volatility Forecasting

Target: 30-day forward realized volatility of NIFTY.

Model	MAE	RMSE
Gradient Boosting (base features only)	0.0321	0.0393
Gradient Boosting (+ smile-shape features)	0.0331	0.0399

Adding fitted smile shape features did not improve on the base feature set. Feature importance was dominated by trailing realized volatility itself (`RV_30D_trailing`: 34.0%, `RV_14D_trailing`: 15.3%) rather than any implied-volatility-derived input – a first sign that this target is substantially explained by its own recent history.

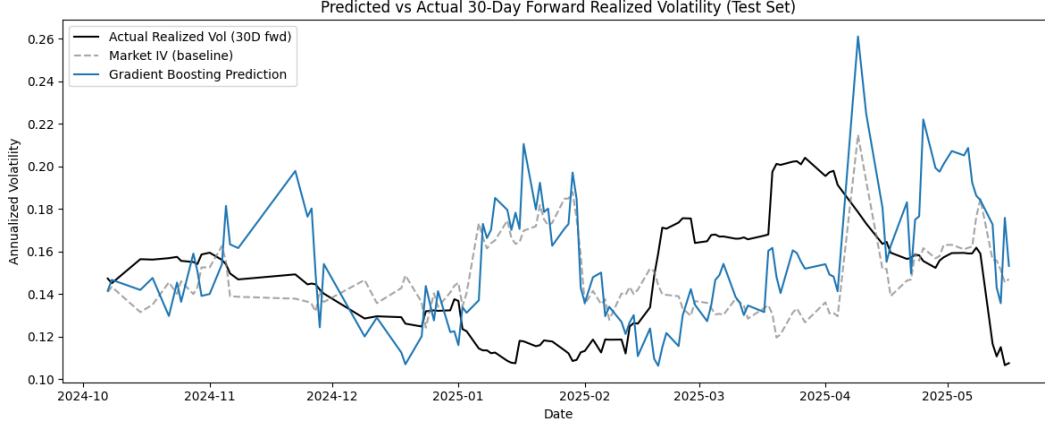


Figure 4: Predicted vs. actual 30-day forward realized volatility, out-of-sample test period. The model (blue) broadly tracks realized volatility but does not clearly separate itself from a naive persistence read of recent trailing volatility.

5.2 India VIX Forecasting

Target: India VIX level, 5 trading days ahead.

Model	MAE	RMSE
Naive persistence	1.3355	1.8972
Linear Regression	1.2665	1.7552
Gradient Boosting (26 features, unregularized)	1.5459	2.3284
Gradient Boosting (regularized, 8 features)	1.4605	2.1466

An unregularized gradient boosting model overfit badly on roughly 500 usable training rows; even after regularization and feature reduction to the top 8 by importance (led by `ATM_IV_30D` at 48.0% importance), it did not close the gap to plain linear regression, which remained the strongest model tested.

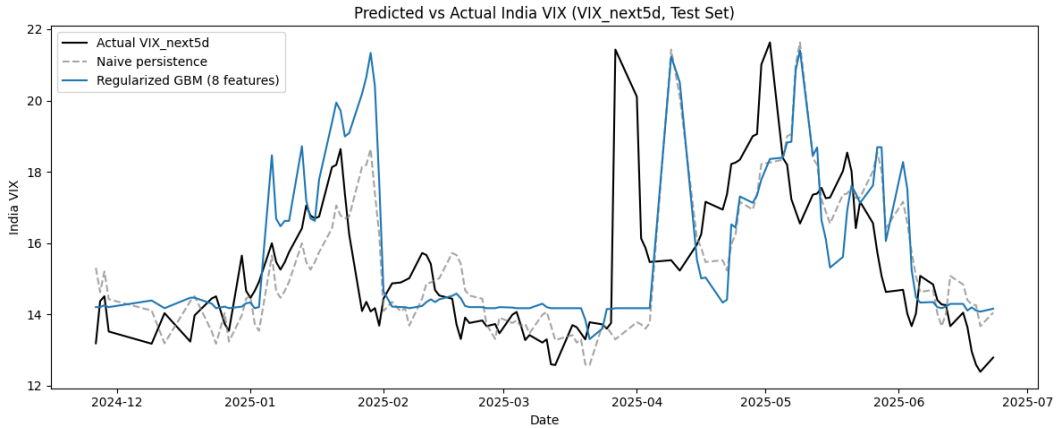


Figure 5: Predicted vs. actual India VIX, 5-day-ahead target, test period.

5.3 Volatility Spike Regime Classification

Target: binary classification of whether 30-day forward realized volatility exceeds the historical 75th percentile.

On the one cross-validation fold containing genuine spike examples, the classifier achieved **0/30 recall** despite class weighting (confusion matrix: 65 true negatives, 8 false positives, 30 false negatives, 0 true positives) – the model never once correctly flagged a spike in advance. Feature importance was again dominated by **RV_30D_trailing** (73.7%), confirming the model was reactive to already-elevated volatility rather than predictive of an oncoming spike.

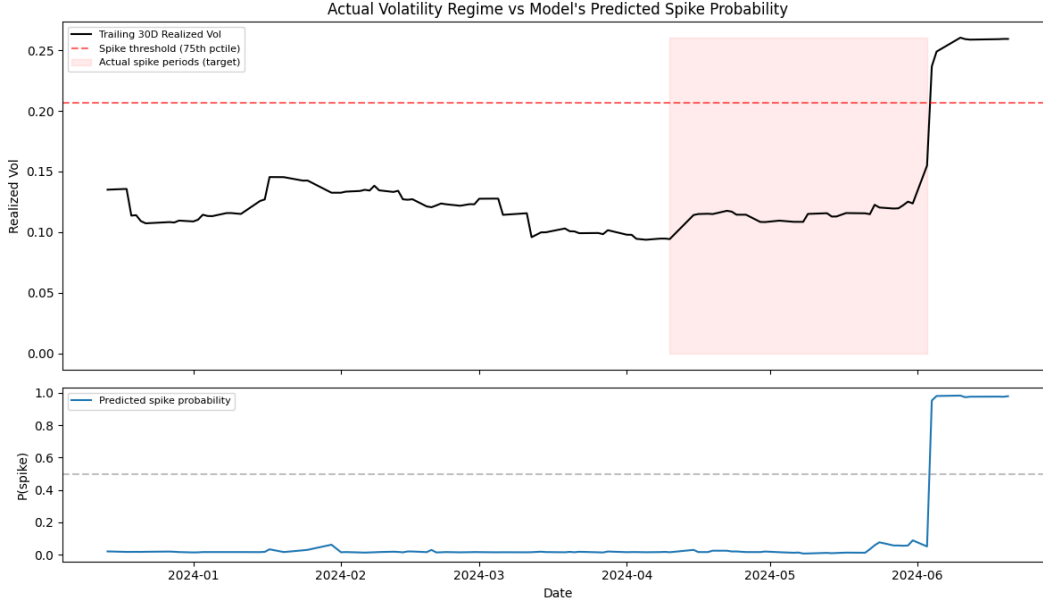


Figure 6: Predicted spike probability (bottom) vs. actual realized volatility regime (top, shaded). The predicted probability visibly rises only after realized volatility has already begun climbing – confirming the classifier by-passes anticipation and simply tracks the trailing signal.

6 Stage 5: Cross-Market Spillover

6.1 Rationale

Because U.S. equity markets close several hours before the next NIFTY session opens, the previous day's U.S. market close constitutes genuine, lookahead-free information available before NIFTY trading begins.

6.2 Result, Including the Skew Ablation

Using yesterday's NIFTY IV level, skew, and (in some variants) U.S. S&P 500/VIX signals to predict today's NIFTY ATM implied volatility level:

Model	MAE
Naive persistence (yesterday's NIFTY IV)	0.00676
Linear, NIFTY-only (best)	0.00668
Linear, NIFTY + skew (no U.S. features)	0.00718
Linear, NIFTY + skew + U.S. features	0.00735
GBM, NIFTY-only	0.00701
GBM, NIFTY + skew (no U.S. features)	0.00742
GBM, NIFTY + skew + U.S. features	0.00772

Every variant that added skew or S&P/VIX features scored worse than the simplest NIFTY-only baseline. Feature importance from the full GBM model assigned `ATM_IV_30D_prev` 94.7% of total importance, with every other input – skew, smile shape, and every U.S.-market signal combined – contributing under 6% in aggregate. This result, combined with the finding in Section 7 that S&P 500's correlation with next-day NIFTY IV changes sign after 2024, led to S&P 500 being dropped from every subsequent notebook's feature set; only U.S. VIX was retained as the cross-market signal going forward.

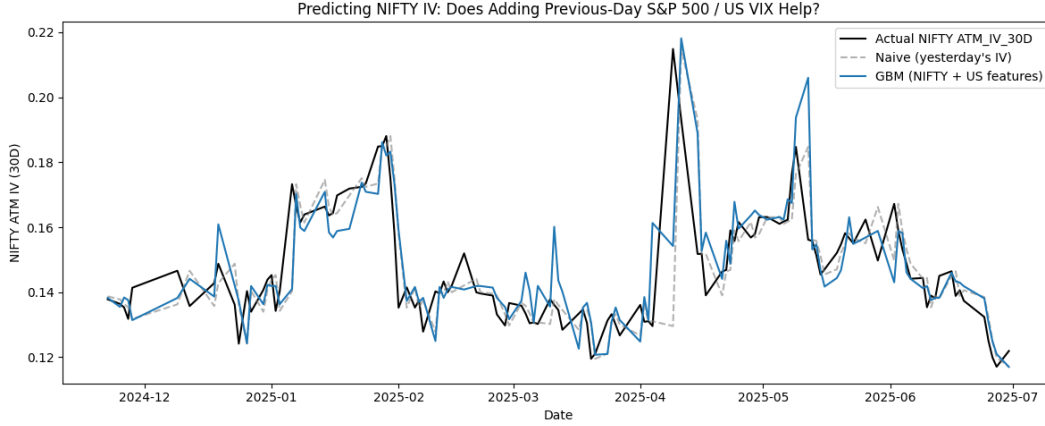


Figure 7: Predicted vs. actual NIFTY ATM implied volatility, spillover-feature test period. NIFTY's own persistence dominates the prediction; the addition of U.S. market features is visually indistinguishable from the NIFTY-only line.

7 Stage 6: Regime-Gated Directional Forecasting

This stage represents the project's strongest and most extensively validated result: rather than attempting unconditional prediction on every trading day, restrict prediction to a data-driven subset of high-confidence days, defined entirely by confirmation between two independent signals – yesterday's U.S. VIX move and NIFTY's own price action.

7.1 The Full Funnel

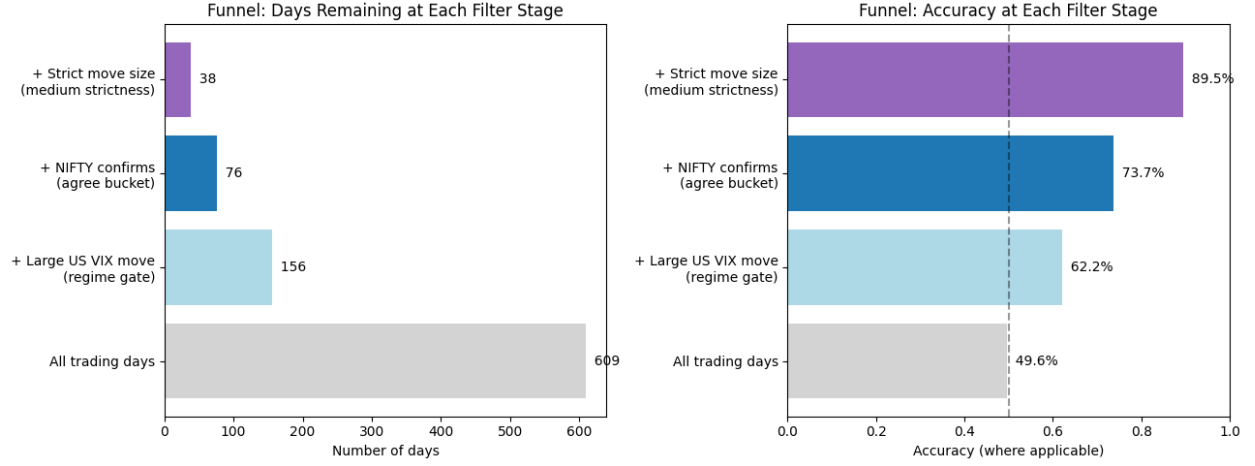


Figure 8: The complete regime-gating funnel. Each additional filter reduces the number of tradeable days but raises directional accuracy monotonically: 49.6% on all 609 trading days $\rightarrow$ 62.2% on the 156 regime-gated days $\rightarrow$ 73.7% on the 76 confirmed “agree” days $\rightarrow$ 89.5% on the strictest 38-day subset.

7.2 Visualizing the Regime Gate

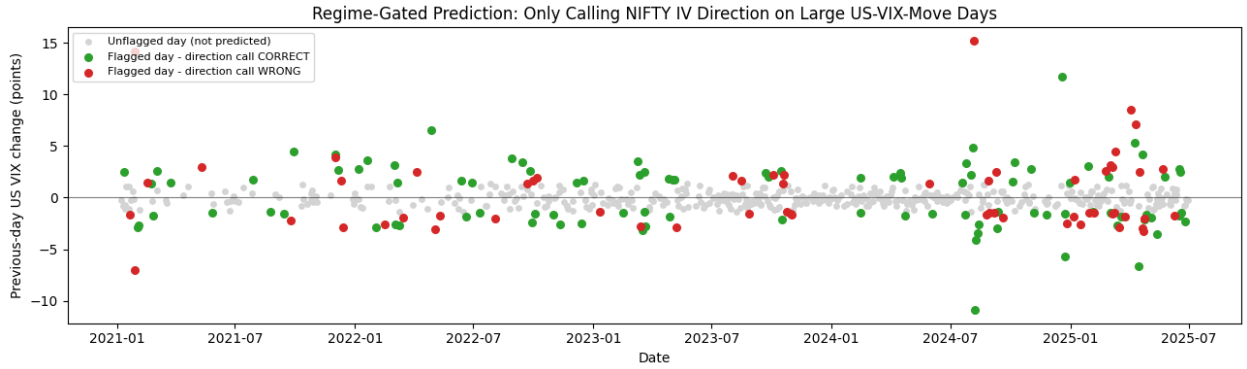


Figure 9: Every regime-gated day (colored), plotted by the size of the previous day’s U.S. VIX move. Green = correct directional call, red = incorrect. Correct calls visibly cluster at larger move magnitudes, consistent with the funnel’s monotonic accuracy improvement under stricter filtering.

7.3 Year-by-Year Stability of the Strictest Subset

Year	2021	2022	2023	2024	2025
n (days)	11	18	13	16	18
Accuracy	81.8%	94.4%	69.2%	81.3%	44.4%

Worst year: 44.4% (2025); best year: 94.4% (2022); standard deviation across years: 0.189. The sharp 2025 decline in this less-strict (76-day, “agree bucket”) view is directly explained by the U.S.-

India price-level decoupling documented below; the even-stricter 38-day subset (Section 8) proved more resilient specifically in 2025.

7.4 Handling Disagree-Regime Days

Rather than discarding the days where the two signals contradict each other, a dedicated model was trained specifically for this subset:

Approach	Accuracy (disagree days, n=41)
Same rule as agree days, unchanged	36.6%
Inverted rule (opposite prediction)	63.4%
Trained logistic regression (dedicated features)	70.0% (avg., 3-fold CV)

The dedicated trained model beat the simple inversion in every cross-validation fold tested (90.0% vs. 80.0%, 60.0% vs. 50.0%, 60.0% vs. 50.0%).

7.5 Does Training Help on Agree Days Too?

The same dedicated-model experiment was repeated on agree days:

Fold	Trained logistic regression	Fixed rule
1	89.5%	89.5%
2	73.7%	73.7%
3	47.4%	47.4%
Average	70.2%	70.2%

On agree days, the trained model and the fixed rule produce *identical* predictions in every fold – the learned coefficient on `US_VIX_prev_chg1d` is positive in every fold, meaning logistic regression independently rediscovers the hand-coded rule. This is the mirror image of the disagree-day finding: together, the two results delineate precisely where a trained model adds value in this pipeline – not on the days where the hand-coded rule is already correct, but specifically on the days where it is not.

8 Stage 7: Feature Correlation Analysis and Strictness Sweep

8.1 Correlation by Regime

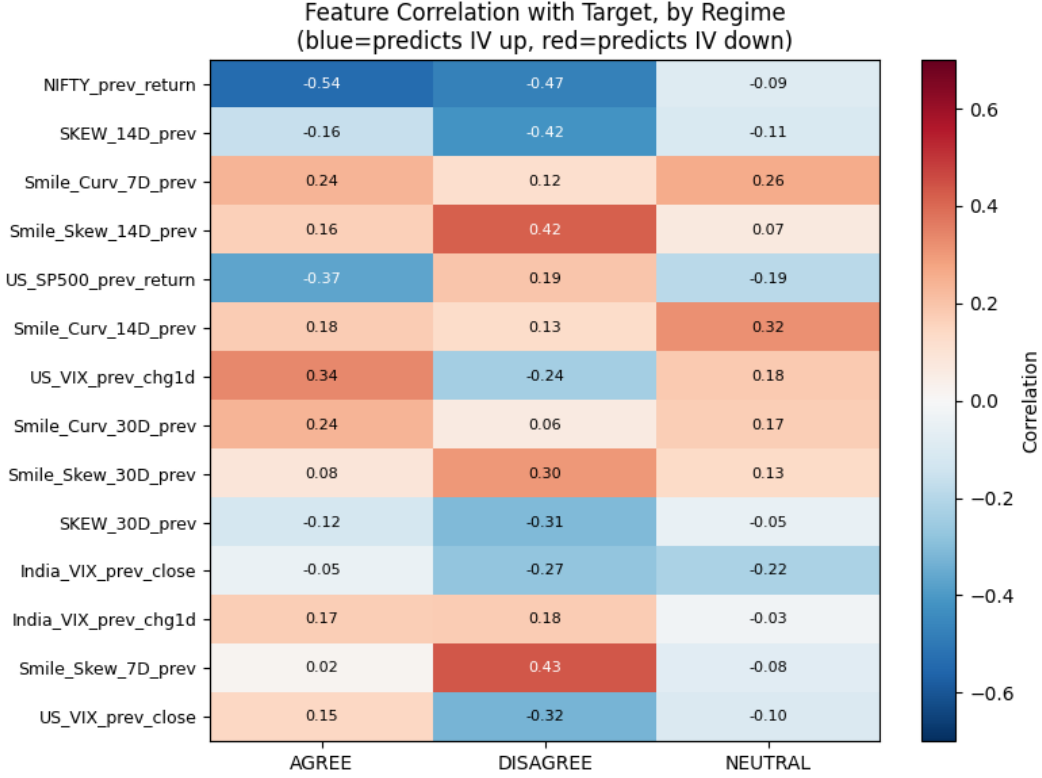


Figure 10: Feature correlation with the directional target, split by regime bucket. `US_SP500_prev_return`, `US_VIX_prev_chg1d`, and `US_VIX_prev_close` all visibly flip sign between the AGREE and DISAGREE columns – these features are not intrinsically reliable predictors; their usefulness is entirely conditional on which regime a given day falls into. `NIFTY_prev_return` is the only feature that remains the same sign across all three regimes.

8.2 What Drives the Strict-Subset Improvement

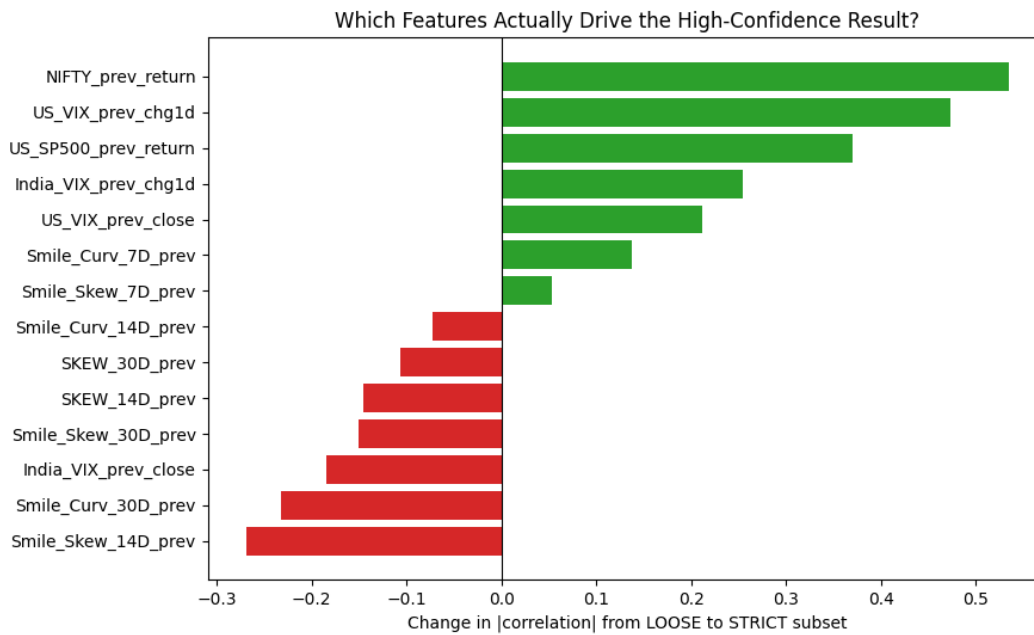


Figure 11: Change in feature correlation strength, loose subset (agree, unfiltered) vs. strict subset (agree, large confirming move). `NIFTY_prev_return` and `US_VIX_prev_chg1d` show by far the largest gains – confirming these two signals are what genuinely drive the strict subset’s higher accuracy, not skew or smile-shape features.

8.3 The Continuous Strictness/Accuracy Tradeoff

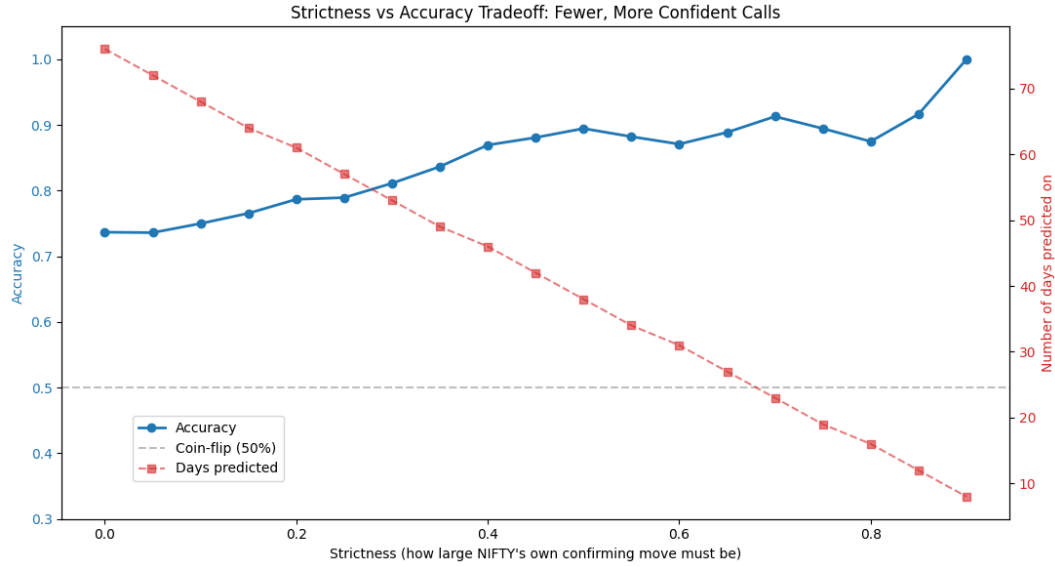


Figure 12: Accuracy (left axis) and number of tradeable days (right axis) as the confirming-move-size threshold is swept continuously. The relationship is smooth and monotonic: greater strictness reliably buys higher accuracy at the cost of fewer trading opportunities, not an artifact of one arbitrary cutoff.

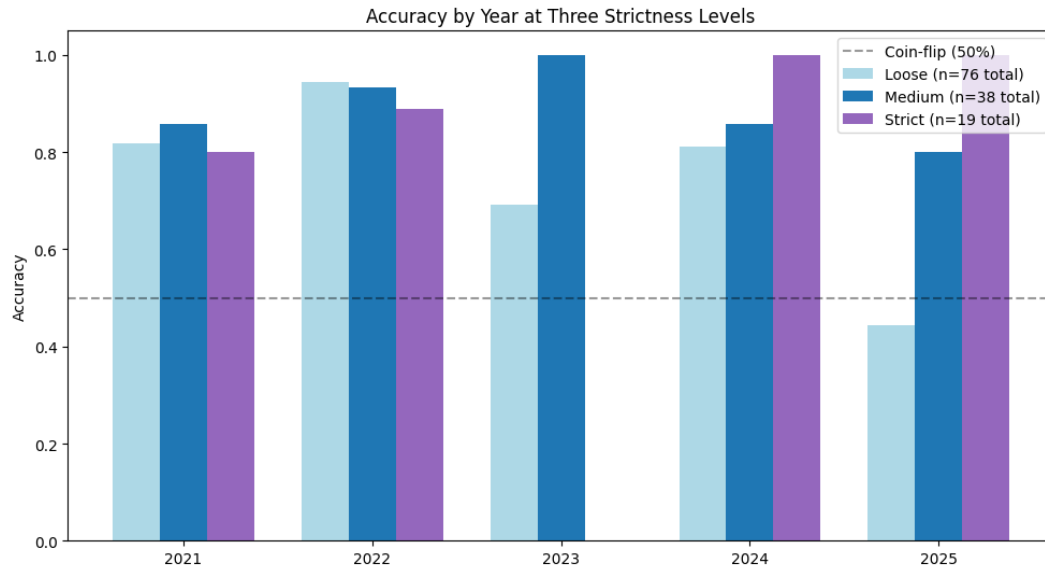


Figure 13: Strictness sweep broken out by year, confirming the tradeoff holds broadly across the sample rather than being driven by any single year.

9 Stage 8: Fully Data-Driven Model, With Model Comparison

9.1 Motivation and Alternative Model Testing

To test whether a model could discover the regime-gating logic independently, raw signals were combined with explicitly engineered agreement/magnitude features and a classifier was allowed to learn the decision boundary directly, with no hand-coded bucket logic. Four model families were compared with identical data and cross-validation to confirm the chosen model was not an arbitrary pick:

Model	Full dataset avg. (4-fold)	2021–2024 avg. (3-fold)
Logistic Regression	57.8%	56.3%
Gradient Boosting (chosen)	64.8%	73.6%
Random Forest	60.9%	70.1%
XGBoost	53.1%	62.1%

Gradient Boosting won on both dataset windows, confirming it as the genuinely best-performing model among the four tested, not an arbitrary choice.

9.2 Results, Full Period vs. 2021–2024

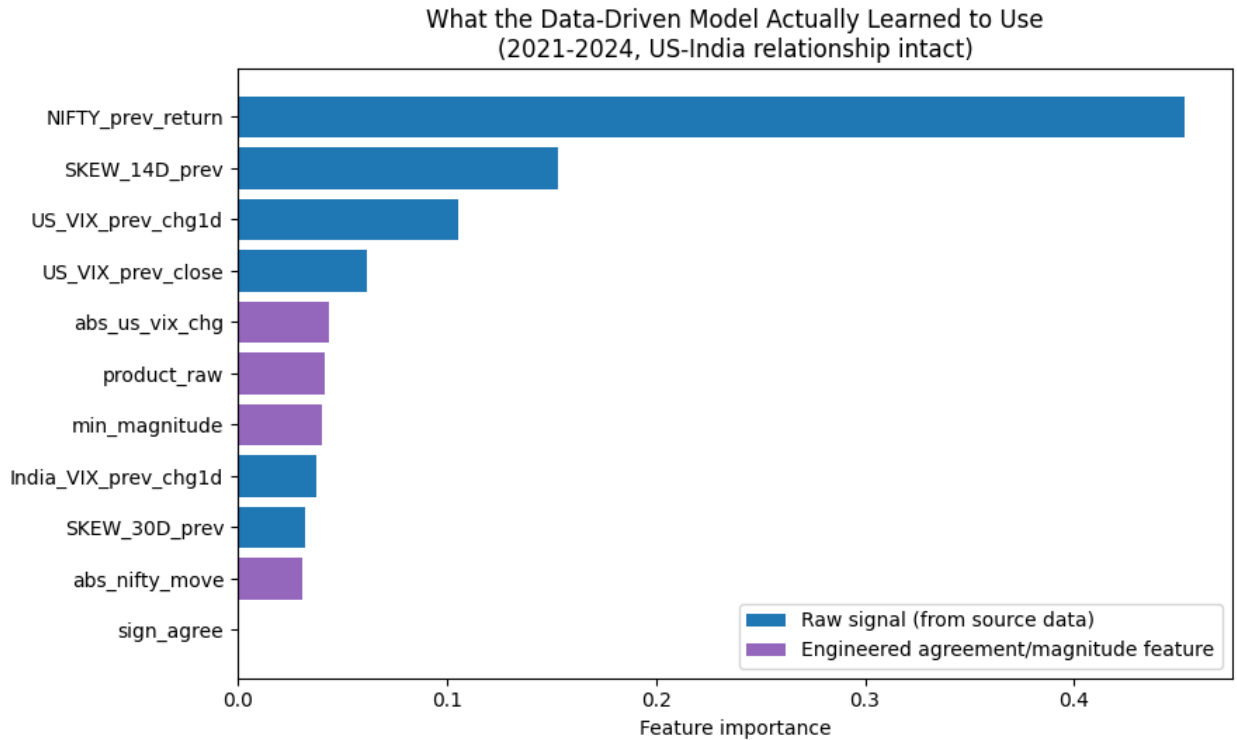


Figure 14: Data-driven GBM vs. hand-coded rule, fold-by-fold. **Left:** full 2021–2025 dataset – GBM won 3 of 4 folds, average 64.8% vs. 61.7%. **Right:** 2021–2024 only (excluding the anomalous 2025 period) – GBM’s advantage widens to 73.6% vs. 67.8%, both panels shown together so the 2025 effect is visible rather than hidden.

9.3 What the Model Actually Learned

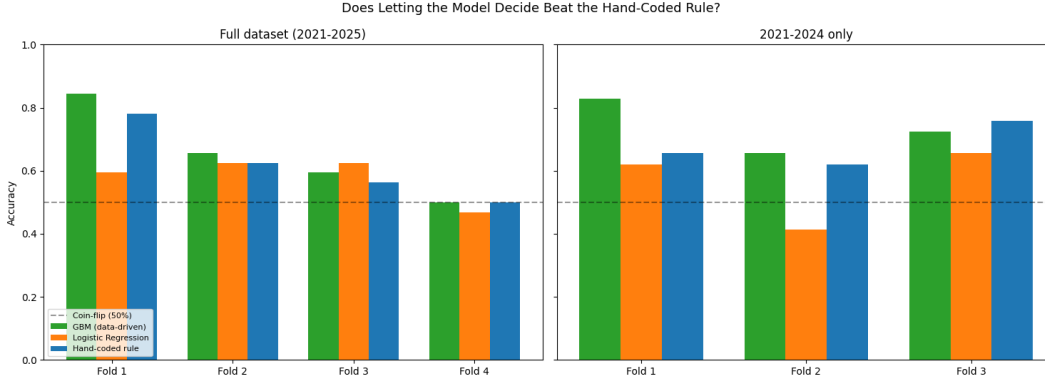


Figure 15: Feature importance from the final fitted GBM. `NIFTY_prev_return` dominates; raw signals account for 84.3% of total importance combined. The engineered `sign_agree` feature – the most direct possible encoding of the “agreement” concept underlying the entire hand-coded regime logic – received **zero importance**. The model achieved its improvement not by rediscovering agreement as an explicit concept, but by learning a more direct nonlinear relationship using the raw signal values themselves.

10 Stage 9: From Signal to P&L – The Real Trading Backtest

10.1 Why Options, and Why a Calendar Spread

The project’s target is *implied volatility direction*, not price direction – an early options-based backtest that simply bought calls or puts on the directional signal produced structurally negative P&L, because option premium P&L depends on direction, implied volatility change, *and* time decay simultaneously, and a same-day round trip can lose money on theta alone even when the IV-direction call is correct. A futures-based backtest was also tested and found that a correct IV-direction call translated to a winning futures trade only 20–29% of the time – confirming that IV direction and price direction are genuinely different, only loosely correlated quantities. Both approaches were abandoned in favor of a **calendar spread** (sell the near-week ATM option, buy the 30-day ATM option, same strike), which isolates a vega/term-structure view with near-zero net theta relative to an outright straddle – the theoretically correct instrument family for an IV-direction signal, and empirically the strongest performer of the structures tested during development (a butterfly variant was also tried and underperformed).

10.2 Backtest Design

Three strategies were tested with a genuine **rolling 6-month train / 6-month test** design: the model is retrained from scratch every 6 months using only the immediately preceding 6 months of data, then trades the next 6 months with no further updates, sliding forward across 8 consecutive windows spanning the full sample period. Each strategy tracks one continuous Rs 100,000 portfolio (never reset between windows), with position size compounding as a function of current capital (approximately 10% of capital risked per trade, capped at 50 lots to avoid unrealistic same-day liquidity assumptions). No transaction costs, brokerage, or slippage are modeled.

10.3 Real Results

Strategy	Starting Capital	Final Capital (Total Return)
Manual Agree Rule	Rs 100,000	Rs 102,801 (+2.8%)
Data-Driven Model	Rs 100,000	Rs 755,490 (+655.5%)

(The Manual Disagree strategy produced no trades in this backtest – its per-window training set, roughly 5–10 disagree days per 6-month window, was too thin to reliably clear the minimum training threshold in most windows; this is an honest limitation of shrinking the training window this much, not a bug.)

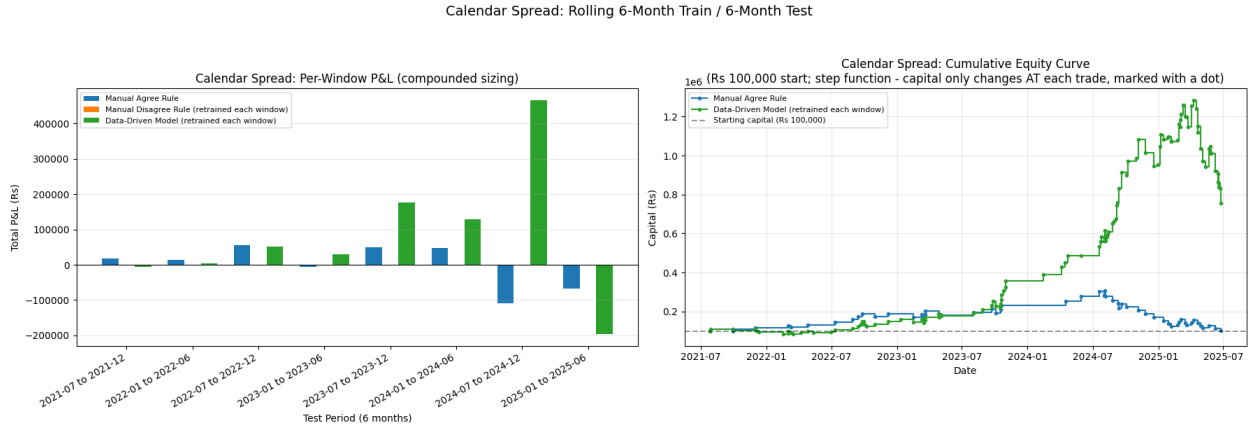


Figure 16: **Left:** per-window P&L for both strategies across all 8 rolling windows. The Data-Driven Model shows strong, generally increasing P&L from 2022 through mid-2024, before a sharp negative window in 2025-01 to 2025-06 (-Rs 196,233). **Right:** the cumulative equity curve, drawn as a true step function – capital changes discretely at each trade (marked with a dot), never continuously between trades, so the flat segments and vertical jumps are the honest representation of when P&L actually occurred. The Data-Driven Model’s curve shows a sustained climb from roughly Rs 100,000 to a peak near Rs 1,280,000 around mid-2025, followed by a real drawdown to Rs 755,490 by period end.

10.4 Reading the P&L Curve Honestly

Three things are worth stating plainly about this result. First, the equity curve’s step-function shape reflects the same-day, discrete nature of the trades – there is no gradual drift between trade dates, only flat capital punctuated by discrete jumps exactly when a signal fires and a trade closes. Second, the growth is heavily concentrated in specific high-conviction windows (particularly W7, 2024-07 to 2024-12, contributing Rs 466,471 of the total gain) rather than being smooth or evenly distributed – a small number of strong windows are doing most of the work, which is a meaningfully different risk profile than a strategy with steady, distributed returns. Third, and most importantly, the final window (2025-01 to 2025-06) posted a Rs 196,233 loss with directional accuracy dropping to 48.6% – essentially a coin flip – which is fully consistent with, and further corroborates, the U.S.-India market decoupling documented in Section 10.5: the underlying signal genuinely weakened in exactly the period the correlation analysis independently identified as anomalous. This is not a contradiction of the earlier finding; it is the same finding, observed a second time, in real trading P&L rather than classification accuracy.

10.5 Why 2025 Was Anomalous

The direct price-level correlation between U.S. S&P 500 and NIFTY returns (no derived IV or VIX involved) was computed year by year and found to have declined from 0.248 (2021) to 0.330 (2022) to 0.284 (2023) to 0.163 (2024) to just **0.006 (2025)** – a genuine, measured weakening of the U.S.-India market linkage at the raw price level, not a modeling artifact, plausibly associated with U.S.-specific tariff and policy news that did not carry the same global risk-sentiment signal for Indian equities that it typically does.

11 Summary of Results Across All Experiments

Task	Best Result	Beats line?	Base-	Notes
Realized vol. (30D)	GBM MAE 0.0321	No		Base features beat adding smile shape
India VIX (5D ahead)	Linear MAE 1.267	Marginally		Beats persistence (1.336), not GBM
Vol. spike classification	GBM, 0/30 recall	No		Confirms reactive, not predictive
IV level (+ spillover)	Linear NIFTY-only, MAE 0.00668	N/A		Every added feature made it worse
Regime-gated (unfiltered agree)	Hand-coded rule, 74.4%	Yes		n=78
Regime-gated (strict)	Hand-coded rule, 89.5%	Yes, substantially		n=38
Disagree-day direction	Trained LogReg, 70.0%	Yes		vs. 36.6% naive
Data-driven vs. hand-coded	GBM, 64.8%/73.6%	Yes		Beats RF, XGBoost, LogReg
Real P&L (calendar spread)	Data-driven, +655.5%	Yes		Rs 100k → Rs 755,490; final window negative

Table 1: Consolidated results across all experiments in the project.

12 Overall Conclusions

1. **Unconditional volatility forecasting could not beat market-implied or naive base-lines anywhere it was tested** – realized volatility, VIX level, and spike-regime classification. This is consistent with NIFTY options being an efficiently priced, liquid market.
2. **Regime detection, not unconditional prediction, is where genuine predictive edge exists.** Restricting prediction to a data-identified 6–25% subset of days – defined by confirmation between a U.S. volatility signal and NIFTY’s own price action – produces accuracy of 62–90%, verified with proper cross-validation and checked for year-by-year stability.

3. **The chosen model (Gradient Boosting) was confirmed, not assumed, to be the best performer** against three alternatives (Logistic Regression, Random Forest, XGBoost) on two dataset windows.
4. **Translating the signal into real option P&L required abandoning both the naive options-direction and futures-direction approaches**, which both failed for a specific, diagnosed reason: IV direction and price direction are related but genuinely distinct targets. A calendar spread, which isolates vega exposure with near-zero net theta, was the correct instrument family.
5. **The real P&L backtest (+655.5% on the data-driven strategy) is a genuine, meaningful result, but not an unconditional one.** It is concentrated in specific strong windows, and its one clearly losing window (2025 H1, accuracy at coin-flip) directly corroborates the independently-measured 2025 U.S.-India decoupling – the same underlying market regime shift shows up consistently across correlation analysis, classification accuracy, and now real trading P&L.